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Literature Review

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Overview of literature review

- Literature review of 13 papers
- Main topics: crack propagation in snow, snow modeling, modeling in more general sense (introduction to different modeling methods used)
- Modeling methods: Material Point Method (MPM) and Discrete Element Method (DEM)

Motivation

- Understanding the different modeling methods and their limitation is important for thesis to understand which methods are suited for what
- Understanding how cracks propagate and which factors influence them is important for risk assessment
- Understanding which factors lead to supershear crack propagation is also important for risk assessment

Modeling Methods

- Two main methods used: MPM and DEM
- DEM described by Cundall and Strack, 1979, material of disks & spheres, contact forces and resulting displacements calculated
- MPM where material is discretized in points with Eulerian background mesh to calculate stresses at points and time-dependent equations, for example in Gaume et al., 2018
- MPM is more favorable for slope-scale simulations because of modelling of grain contacts
- Other models for snow such as SNOWPACK (Bartelt & Lehning, 2002) and CROCUS (Brun et al., 1992) model snowpack response to external factors

Avalanche Release - General Crack Propagation

- All papers agree on avalanche formation by loading - weak layer failure
- Some papers discuss anticrack propagation (closure of cracks formed) like eg Gaume et al., 2018
- Others state shear cracks to be the main failure mode
- Bergfeld et al., 2025 combine the two main hypotheses: anticrack early in avalanche propagation, shearing later

Factors influencing crack propagation

- Slope angle: higher slope angle generally increases crack speed and can trigger transition from sub-Rayleigh to supershear; it can also reduce the critical crack length needed for release (Bobillier et al., 2024; Gaume et al., 2017; Trottet et al., 2022)
- Slab stiffness (elastic modulus): stiffer slabs lead to lower normalized crack speed in some DEM results, but it tends to increase critical crack length (more crack growth needed before release) (Bobillier et al., 2024; Gaume et al., 2017)
- Weak layer strength & elastic modulus: higher weak-layer elastic modulus can increase crack speed; stronger weak layers generally require more energy/longer cracks for release (Bobillier et al., 2024; Gaume et al., 2015; Gaume et al., 2017)

Factors influencing crack propagation (continued)

- Slab density: higher slab density is associated with faster propagation in some models, while also reducing critical crack length for release (Gaume et al., 2015; Gaume et al., 2017)
- Slab thickness: thicker slabs can increase propagation speed in DEM studies; in critical-crack formulations, greater slab thickness lowers critical crack length (Gaume et al., 2015; Gaume et al., 2017)
- Weak layer thickness: thinner weak layers can lead to faster crack propagation and lower critical crack length for release (Gaume et al., 2015; Gaume et al., 2017)
- Snow structure and loading rate: sintering and lower loading rates can lead to more stable snowpacks, which means higher critical crack lenths (Mulak & Gaume, 2019)

Deformation stages

- After loading, multiple stages of deformation
- loading response by an elastic regime, a drop in pressure and shear stress, volumetric collapse of the weak layer and lastly dense packing (Gaume et al., 2018)
- Slab bending induced by weak layer collapse at bottom of snowpack (Bergfeld et al., 2025)
- Deformation because of slab tension at top of snow (Bergfeld et al., 2025)

Crack propagation direction

- Direction of crack propagation dependent on simulation scale (Trottet et al., 2022)
- Small-scale: propagation upslope (remote triggering), because collapse depth for bending limited by weak layer collapse (Trottet et al., 2022)
- Large-scale: downslope, because propagation speed can increase because of slab tension (Trottet et al., 2022)

Supershear crack propagation I

- Supershear crack propagation when cracks propagate above elastic wave speed in snow (Trottet et al., 2022)
- Snow properties influence if supershear speed is reached
- High slab tensile strength: supershear cracks (Trottet et al., 2022)
- Increased slab tension in steep terrains: high speeds (possible supershear) (Trottet et al., 2022)
- Slope angle dependance: at below 30°, sub-rayleigh, at above 30°: supershear (Bobillier et al., 2024)
- Also property dependent (weak layer elastic modulus snow elastic modulus) (Bobillier et al., 2024)
- Generally: the "longer" cracks can propagate before breakoff, the higher the speed they can reach (direct failure in weak slabs, no failure in very strong slabs)
- Early stages of crack propagation are anticrack, later stages supershear (Bergfeld et al., 2025)

Bibliography I

Bartelt, P., & Lehning, M. (2002).A physical SNOWPACK model for the Swiss avalanche warning: Part I: Numerical model. *Cold Regions Science and Technology*, *35*(3), 123–145. https://doi.org/10.1016/S0165-232X(02)00074-5

Bergfeld, B., Gaume, J., Bobillier, G., Pellet, A., Schweizer, J., & van Herwijnen, A. (2025).Signatures of the sub-Rayleigh to supershear fracture transition in snow avalanche experiments. *Nature Communications*, *16*(1), 11153. https://doi.org/10.1038/s41467-025-65825-6

Bobillier, G., Bergfeld, B., Dual, J., Gaume, J., van Herwijnen, A., & Schweizer, J. (2024).Numerical investigation of crack propagation regimes in snow fracture experiments. *Granular Matter*, *26*(3), 58. https://doi.org/10.1007/s10035-024-01423-5

Brun, E., David, P., Sudul, M., & Brunot, G. (1992).A numerical model to simulate snow-cover stratigraphy for operational avalanche forecasting. *Journal of Glaciology*, *38*(128), 13–22. https://doi.org/10.3189/S0022143000009552

Bibliography II

Cundall, P. A., & Strack, O. D. L. (1979). A discrete numerical model for granular assemblies. *Géotechnique*, *29*(1), 47–65. https://doi.org/10.1680/geot.1979.29.1.47

Gaume, J., Gast, T., Teran, J., van Herwijnen, A., & Jiang, C. (2018). Dynamic anticrack propagation in snow. *Nature Communications*, *9*(1), 3047. https://doi.org/10.1038/s41467-018-05181-w

Gaume, J., van Herwijnen, A., Chambon, G., Birkeland, K. W., & Schweizer, J. (2015). Modeling of crack propagation in weak snowpack layers using the discrete element method. *The Cryosphere*, *9*(5), 1915–1932. https://doi.org/10.5194/tc-9-1915-2015

Gaume, J., van Herwijnen, A., Chambon, G., Wever, N., & Schweizer, J. (2017). Snow fracture in relation to slab avalanche release: Critical state for the onset of crack propagation. *The Cryosphere*, *11*(1), 217–228. https://doi.org/10.5194/tc-11-217-2017

Bibliography III

Mulak, D., & Gaume, J. (2019). Numerical investigation of the mixed-mode failure of snow. *Computational Particle Mechanics*, *6*(3), 439–447. https://doi.org/10.1007/s40571-019-00224-5

Trottet, B., Simenhois, R., Bobillier, G., Bergfeld, B., van Herwijnen, A., Jiang, C., & Gaume, J. (2022). Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches. *Nature Physics*, *18*(9), 1094–1098. https://doi.org/10.1038/s41567-022-01662-4

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